

ENGR1050: Lecture 13

Differential equations and numerical methods

October 8, 2025



Agenda

- 5 minute crash course on differential equations
- Together: build up a class to solve ODEs using Euler's methods
- Independent practice: use your class to solve a few ODEs

Types of Equations

- **Nonlinear equations:**

- Systems of equations that must be equal.
- $F(x_1, x_2, \dots) = 0$
- Example: $x_1^2 + x_2^2 - 1 = 0$ (circle equation).

- **Ordinary Differential Equations (ODEs):**

- Involve derivatives with respect to a single variable (e.g., time).
- $F(t, y, y', y'', \dots) = 0$
- Example: $\frac{dy}{dt} = -ky$ (exponential decay).
- Example: $m\ddot{x} = F(x)$ (Newton's second law).

- **Partial Differential Equations (PDEs):**

- Involve derivatives with respect to multiple variables, typically mixing space and time
- Example: $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$. (heat equation).

What is an ODE?

Definition

An n^{th} -order ordinary differential equation involves an unknown function $y(t)$ and its derivatives up to order n , where n is the highest derivative present:

$$F(t, y, y', \dots, y^{(n)}) = 0.$$

Equivalently, in explicit form:

$$y^{(n)} = f(t, y, y', \dots, y^{(n-1)}).$$

Example: A ball dropped from height h with air resistance proportional to velocity:

$$m \frac{d^2 y}{dt^2} = -mg - k \frac{dy}{dt} \quad \Rightarrow \quad \frac{d^2 y}{dt^2} = -g - \frac{k}{m} \frac{dy}{dt}$$

is a *second-order* ODE since the highest derivative is the second derivative of position y with respect to time t .

Initial value problems

Definition

An initial value problem specifies an ODE together with values of the unknown at an initial time t_0 . For an n^{th} -order ODE:

$$y^{(n)} = f(t, y, y', \dots, y^{(n-1)}).$$

with initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_1, \quad \dots, \quad y^{(n-1)}(t_0) = y_{n-1}.$$

Example: For the falling ball ODE above, to fully specify the trajectory of the ball we must specify the initial height and velocity through the *initial conditions*:

$$y(0) = h, \quad y'(0) = 0.$$

Why do we need initial conditions?

- Lets think about exponential decay:

$$\frac{dy}{dt} = -ky \quad (\text{rate proportional to magnitude of population})$$

- If we integrate both sides we obtain:

$$\int \frac{1}{y} dy = -k \int dt \quad \Rightarrow \quad \ln |y| = -kt + C.$$

- Exponentiating both sides gives:

$$|y| = e^{-kt+C} = e^C e^{-kt}.$$

- Letting $C' = e^C$ (a positive constant), we have:

$$|y| = C' e^{-kt}.$$

- Thus, the general solution is:

$$y(t) = Ce^{-kt},$$

where C is specified by the initial conditions $C = y(t_0)$.

How we'll think about ODEs in ENGR1050

You'll take a full course on differential equations later in your studies, but for this class we will either:

- Guess and check solutions to simple ODEs
- Use numerical methods to approximate solutions to more complex ODEs

Example of how to guess and check

Example: Verify that $y(t) = e^{-2t}$ is a solution to the ODE:

$$\frac{dy}{dt} = -2y.$$

- Compute the derivative of $y(t)$:

$$\frac{dy}{dt} = -2e^{-2t}.$$

- Substitute $y(t)$ into the right side of the ODE:

$$-2y = -2e^{-2t}.$$

- Since both sides are equal, $y(t) = e^{-2t}$ is indeed a solution to the ODE.

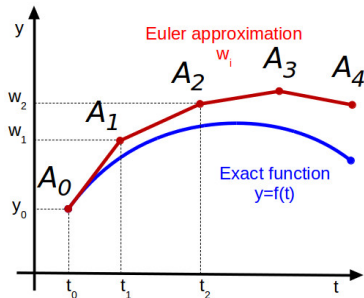
Numerical solution of ODEs

Many ODEs cannot be solved analytically, so we use numerical methods to approximate solutions. The simplest is the Explicit Euler method.

Definition

For the IVP $y'(t) = f(t, y)$ with $y(t_0) = y_0$ and step size $h > 0$, the Starting from t_0, y_0 , Explicit Euler scheme updates via

$$t_{n+1} = t_n + h, \quad y_{n+1} = y_n + h f(t_n, y_n),$$



How will we check that a solution is reasonable?

In the next homework we will use the code we built today to solve a pile of ODEs. The art of this is to cook up some qualitative feature we know our solution has to have and check that the numerical solution matches it.

Example: For the falling ball with air resistance, we know that:

- The ball should never go below ground level (i.e., $y(t) \geq 0$ for all t).
- The velocity should approach a terminal velocity as time goes on.

$$\ddot{y} = -g - \frac{k}{m}\dot{y}$$

At steady state:

$$0 = -g - \frac{k}{m}v_{term} \implies v_{term} = -\frac{mg}{k}$$

In the homework I'll guide you through some of these checks.

Today's exercise

- Together, we will build a Python class to solve ODEs using the Explicit Euler method.
- The class will take as input:
 - Initial conditions t_0 and y_0
 - Final time t_f to solve to and number of steps N
- The class will have two functions:
 - $f(t,y)$ to define the ODE $y' = f(t,y)$
 - `solve()` to perform N steps and return the solution as a numpy array
- You will use this class to solve a few example ODEs on your own.